Universiti Kebangsaan Malaysia
The National University of Malaysia

NARATIF BAHARU UKM
UNIVERSITI WATAN KITA

FAKULTI SAINS & TEKNOLOGI

DATA ANALYSIS WITH R

Noratiqah Mohd Ariff

DATA ANALYSIS: HYPOTHESIS TESTING

Steps for Hypothesis Testing

1. Define the null and alternative hypotheses; i.e. H_0 and H_1 respectively.
2. Choose the suitable test and select α.
3. Compute the test statistic.
4. Make a statistical decision based on critical value or the p-value, i.e.
   i. Reject H_0 and accept H_1 if p-value is smaller than significance level. This indicates that there is evidence against the null hypothesis.
   ii. Fail to reject H_0 if p-value is bigger than significance level. This indicates that there is not enough evidence against the null hypothesis.

Hypothesis Testing for Means

Comparison of MEANS	Degrees of Freedom	Application	Assumptions	Test Statistic
One Sample Z-Test	Not Applicable	Testing the difference of a sample mean, x-bar, with a known population mean, μ (fixed mean, historical mean, or targeted mean)	Normal distribution Known population σ.	$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$
One Sample t-test	n-1	Testing the difference of one sample mean, x-bar, with a known population mean, μ (fixed mean, historical mean, or targeted mean)	Normal distribution Population standard deviation, σ, is unknown.	$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$
Two Sample t-test	$n_1 + n_2 - 2$	Testing difference of two sample means when population variances unknown but <u>considered equal</u>	Normal Distribution Requires standard pooled deviation calculation, s_p	$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$
Paired t-test	n - 1	Testing two sample means when their respective population standard deviations are unknown but considered equal. Data recorded in pairs and each pair has a difference, d.	Normal Distribution Two dependent samples Always two-tailed test S_d = standard deviation of the differences of all samples	$t = \frac{\bar{d}\sqrt{n}}{s_d}$
One-Way ANOVA	$n_1 - 1$ & $n_2 - 1$	Testing the difference of three or more population means	Normal Distribution s_1^2 and s_2^2 represent sample variances	$F = \frac{(s_1)^2}{(s_2)^2}$

Z-test in R

- We can do z-test without any package using `qnorm()` to find the value for $z_{\alpha/2}$.
- Alternatively, we can apply the `z.test()` function in the `TeachingDemos` package. It is not a core R package and must be installed and loaded into the workspace beforehand.

```
z.test(x,mu,stdev,alternative=c("two.sided","less","greater"),
conf.level)
```

$H_0: \mu = \text{mu}$
Default mu=0

$H_1: \mu \neq \text{mu}$

$H_1: \mu < \text{mu}$

$H_1: \mu > \text{mu}$

Default
conf.level=0.95

```
library(TeachingDemos)
x<-rnorm(100,mean=2,sd=0.5)
z.test(x, mu=2, stdev=0.5, alternative="two.sided")
```

One Sample t-test in R

- The `t.test()` function will perform a test for a population mean of a small (normally distributed) sample.

```
t.test(x,mu,alternative=c("two.sided","less","greater"),
conf.level)
```

$H_0: \mu = \text{mu}$
Default mu=0

$H_1: \mu \neq \text{mu}$

$H_1: \mu < \text{mu}$

$H_1: \mu > \text{mu}$

Default
conf.level=0.95

```
x<-rnorm(20,mean=1,sd=0.5)
t.test(x, mu=2, alternative="less")
```

Two Sample t-test in R

- The `t.test()` function will also perform a test the difference in population means for two independent small (normally distributed) samples.

```
t.test(x,y,mu,alternative=c("two.sided","less","greater"),
conf.level,var.equal)
```

$H_0: \mu_x - \mu_y = \text{mu}$
Default mu=0

$H_1: \mu_x - \mu_y \neq \text{mu}$

$H_1: \mu_x - \mu_y < \text{mu}$

$H_1: \mu_x - \mu_y > \text{mu}$

Default
conf.level=0.95

Default
var.equal=FALSE

```
x<-c(1.3,1.5,1.2,1.7,1.3)
y<-c(1.6,1.7,1.8,1.6,1.5)
t.test(x,y,mu=0,alternative="less",var.equal=TRUE)
```

Paired Sample t-test in R

- The `t.test()` function will also perform a test the difference in population means for two paired samples (differences normally distributed).

```
t.test(x,y,mu,alternative=c("two.sided","less","greater"),
conf.level,paired=T)
```

$H_0: \mu_x - \mu_y = \text{mu}$
Default mu=0

$H_1: \mu_x - \mu_y \neq \text{mu}$

$H_1: \mu_x - \mu_y < \text{mu}$

$H_1: \mu_x - \mu_y > \text{mu}$

Default conf.level=0.95

Need this argument to perform the paired t-test

```
x<-c(1.3,1.5,1.2,1.7,1.3)
y<-c(1.6,1.7,1.8,1.6,1.5)
t.test(x,y,mu=0,alternative="less",var.equal=TRUE,paired=TRUE)
```

Basic Framework of ANOVA

- ANOVA, or Analysis of Variance, is a test used to determine differences between research results from three or more unrelated samples or groups.
- Want to study the effect of one or more qualitative variables on a quantitative outcome variable.
- Qualitative variables are referred to as factors.
- Characteristics that differentiates factors are referred to as levels.
- The one-way is because each value is classified in exactly one way.
- Assume (1) normality, (2) equal variance, and (3) independent samples.

ANOVA

- The null hypothesis is that the means are all equal:

$$H_0\colon\ \mu_1 = \mu_2 = \cdots = \mu_k$$

- The alternative hypothesis is that at least one of the means is different:

$$H_1\colon\ \mu_i \neq \mu_j \ \ (\text{for some } i \text{ and } j)$$

- Basic idea is to partition total variation of the data into two sources
    1. Variation within levels (groups)
    2. Variation between levels (groups)
- If H_0 is true, the standardized variances are equal to one another.

ANOVA

- Variation is simply average squared deviations from the mean:

$$SST = SST_G + SST_E$$

- SST = Sum of squared deviations about the grand mean across all N observations.
- SST_G= Sum of squared deviations for each group mean about the grand mean.
- SST_E = Sum of squared deviations for all observations within each group from that group mean, summed across all groups.
- To make the sum of squares comparable, we divide each one by their associated degrees of freedom, df: $df_T = N - 1, df_G = k - 1, df_E = N - k$
- Hence, we will get mean squares; $MST_G = {}^{SST_G}/_{k-1}$ and $MST_E = {}^{SST_E}/_{N-k}$
- The test statistic is the ratio of group and error mean squares: $F = {}^{MST_G}/_{MST_E}$

ANOVA

- If H_0 is true MST_G and MST_E are equal.
- Critical value for rejection region is $F_{\alpha,k-1,N-k}$
- ANOVA table:

Source of Variation	df	Sum of Squares	Mean Squares	F
Group	$k-1$	SST_G	MST_G	MST_G / MST_E
Error	$N-k$	SST_E	MST_E	
Total	$N-1$	SST		

ANOVA in R

- Apply the `aov()` function.
- One-way ANOVA:

```
library(lattice)
one.way <- aov(yield ~ variety, data = barley)
summary(one.way)
```

- Two-way ANOVA:

```
two.way <- aov(yield ~ variety+site, data = barley)
summary(two.way)
```

- Two-way ANOVA with interaction effect:

```
interaction <- aov(yield ~ variety*site, data = barley)
summary(interaction)
```

Example: Genotyping of a single SNP

- Our data:
  - AA: 82, 83, 97
  - AG: 83, 78, 68
  - GG: 38, 59, 55

```
SNP_type<-rep(c("AA","AG","GG"),c(3,3,3))
SNP_value<-c(82,83,97,83,78,68,38,59,55)
SNP<-data.frame(SNP_type,SNP_value)
SNP.aov<-aov(SNP_value~SNP_type,data=SNP)
summary(SNP.aov)
```

Example: Genotyping of a single SNP

```
            Df Sum Sq Mean Sq F value  Pr(>F)
SNP_type     2   2124  1062.1   12.59 0.00712 **
Residuals    6    506    84.3
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hypothesis Testing for Proportions and Standard Deviations

	1 Sample	2 Samples
Standard Deviation	$H_0: \sigma = \sigma_0$ $H_A: \sigma \neq \sigma_0$ Chi-Square Test	$H_0: \sigma_1 = \sigma_2$ $H_A: \sigma_1 \neq \sigma_2$ F-Test
Proportion	$H_0: \Pi = \Pi_0$ $H_A: \Pi \neq \Pi_0$ Test for One Proportion	$H_0: \Pi_1 = \Pi_2$ $H_A: \Pi_1 \neq \Pi_2$ Test for Two Proportions

One Sample Test for Proportions in R

- The `prop.test()` function will perform a test for a population proportion for a large sample (at least 10 successes and failures).

Number of successes

Number of trials

$H_0: \pi = \mathrm{p}$
Default p=0.5

```
prop.test(x,n,p,alternative=c("two.sided","less","greater"),
conf.level,correct=FALSE)
```

$H_1: \pi \neq \mathrm{p}$

$H_1: \pi < \mathrm{p}$

$H_1: \pi > \mathrm{p}$

Default conf.level=0.95

If this is TRUE, the function will use a continuity correction in the calculation

```
prop.test(52,100,p=0.4,alternative="greater")
```

Two Sample Test for Proportions in R

- The `prop.test()` function will also perform a test for the difference in population proportions for two independent large samples (at least 10 successes and failures).

Vector of length 2: Number of successes

Vector of length 2: Number of trials

$H_0: \pi_x - \pi_y = \mathrm{p}$
No argument p because p=0

```
prop.test(x,n,alternative=c("two.sided","less","greater"),
conf.level,correct=FALSE)
```

$H_1: \pi_x - \pi_y \neq 0$

$H_1: \pi_x - \pi_y < 0$

$H_1: \pi_x - \pi_y > 0$

Default conf.level=0.95

If this is TRUE, the function will use a continuity correction in the calculation

```
prop.test(c(132,135),c(400,390),alternative="two.sided",
correct=FALSE)
```

Chi-Square Test for Variance in R

- We can apply the `varTest()` function in the `EnvStats` package. It is not a core R package and must be installed and loaded into the workspace beforehand.
- It will perform the chi-square test for variance in R.

```
varTest(x,sigma.squared,alternative=c("two.sided","less","greater"),
conf.level)
```

$H_0: \sigma^2 =$ sigma.squared
Default sigma.squared = 1

$H_1: \sigma^2 \neq$ sigma.squared

$H_1: \sigma^2 >$ sigma.squared

$H_1: \sigma^2 <$ sigma.squared

Default
conf.level=0.95

```
library(EnvStats)
data<-c(12.43, 11.71, 14.41, 11.05, 9.53, 11.66,
9.33,11.71,14.35,13.81)
varTest(data,alternative="greater",sigma.squared = 2.25)
```

Equality of Variance Test in R

- The `var.test()` function will test for the equality of variance for samples from two normal populations.

$H_0: \sigma_x^2 / \sigma_y^2 = \text{ratio}$

Default ratio = 1

```
var.test(x,y,ratio,alternative=c("two.sided","less","greater"),
conf.level)
```

Default conf.level=0.95

$H_1: \sigma_x^2 / \sigma_y^2 \neq \text{ratio}$

$H_1: \sigma_x^2 / \sigma_y^2 < \text{ratio}$

$H_1: \sigma_x^2 / \sigma_y^2 > \text{ratio}$

```
x<-c(1.3,1.5,1.2,1.7,1.3)
y<-c(1.6,1.7,1.8,1.6,1.5)
var.test(x,y,alternative="two.sided")
```

DATA ANALYSIS: LINEAR REGRESSION

Linear Regression

- It is a statistical method that is used to:
  - Study relationship among variables
  - Forecast/predict value of variable interest
- There are two types of variables:
  - Dependent variable, y (response variable)
  - Independent variable, x (predictor/regressor variable)
- Regression only study the relationship between the quantitative variables
- For the qualitative variables (eg: colour, race, ...), it needs to be transformed to the numerical values (in quantitative value)

Steps for Regression Analysis

1. The data collected for each variables are used to see the pattern of the relationship/ model existed, whether it is linear/ non-linear relationship.
2. Predict the parameters in the model using linear regression.
3. Next, check for the suitability of the model build based on the data collected. This is to determine whether we should modify or just accept the model we build.
4. Make predictions using the model.

Simple Linear Regression

- Simple linear regression consists of a single regressor, predictor or independent variable X and a dependent or response variable Y.
- The Equation of straight line relating this two variables is: $y_i = \beta_0 + \beta_1 x_i$
- This equation represents the exact relation between x and y; where each point of (x,y) should lie on the straight line.
- Since data points do not fall exactly on a straight line, so we need to modify the Equation by adding the error term in the model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Simple Linear Regression

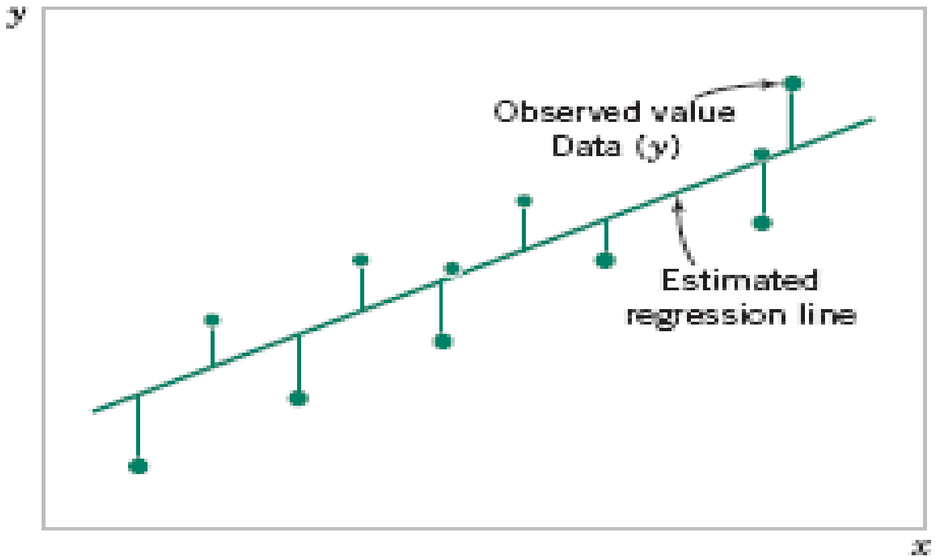
- The difference between the observed value of y (y_i) and the straight line is the error:

$$y_i - (\beta_0 + \beta_1 x_i) = \varepsilon_i$$

- The error is a statistical error (a random variable that accounts for the failure of the model to fit the data exactly).
- The error may made up of the effects of other variables.
- The true regression model is a line of mean values: $\mu_{Y|x} = \beta_0 + \beta_1 x$ where β_1 can be interpreted as the change in the mean of Y for a unit change in x.

Simple Linear Regression

- Estimate the parameter of β_0 and β_1 based on the n pairs of observations (x_1, y_1), (x_2, y_2), …, (x_n, y_n).
- Target! – to get the best straight line which all/ mostly points are lie on this line.
- In regression, we only focus on the error in the y variable. The error in x can be ignored, since x can be fixed/ control by the researcher.



Simple Linear Regression

- Also, the variability of Y at a particular value of x is determined by the error variance, σ^2.
- This implies there is a distribution of Y-values at each x and that the variance of this distribution is the same at each x.
- We assume that each observation, y_i, can be described by the model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Hypothesis Test in Simple Linear Regression

- We can use *t*-test to test the significance of β_0 and β_1.
- The null and alternative hypotheses are:

  $H_0: \beta_0 = \beta_{0,0}$ vs $H_1: \beta_0 \neq \beta_{0,0}$ and $H_0: \beta_1 = \beta_{1,0}$ vs $H_1: \beta_1 \neq \beta_{1,0}$ respectively.
- An important special case: $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$. These hypotheses relate to the significance of regression. Failure to reject H_0, is equivalent to concluding that there is no proof that there is no linear relationship between X and Y.
- *F*-test for regression model (ANOVA test for regression) can also be used to test the significance of regression model.

Coefficient of Determination (R^2)

- Coefficient of determination (R^2) is often used to judge the adequacy of a regression model and is one of the measure of association between X and Y.
- It measures the proportionate reduction in the total variation of Y associated with the introduction of X. Hence, the proportion of total variation in Y which have been explained by X.
- The value of R^2 is within [0,1] where R^2 closer to 1 means that the percentage of variation in Y that is explained by the variation in X is higher.

Coefficient of Determination (R^2)

- Low value of R^2 means that:
  - The current independent random variable of the model does not give an accurate prediction for Y, need to find other suitable X.
  - One independent random variable maybe inadequate, possibly need to add more X.
- High value of R^2 does not necessarily imply that model will provide accurate predictions of future observations and did not measure the appropriateness of the model.

Linear Regression in R

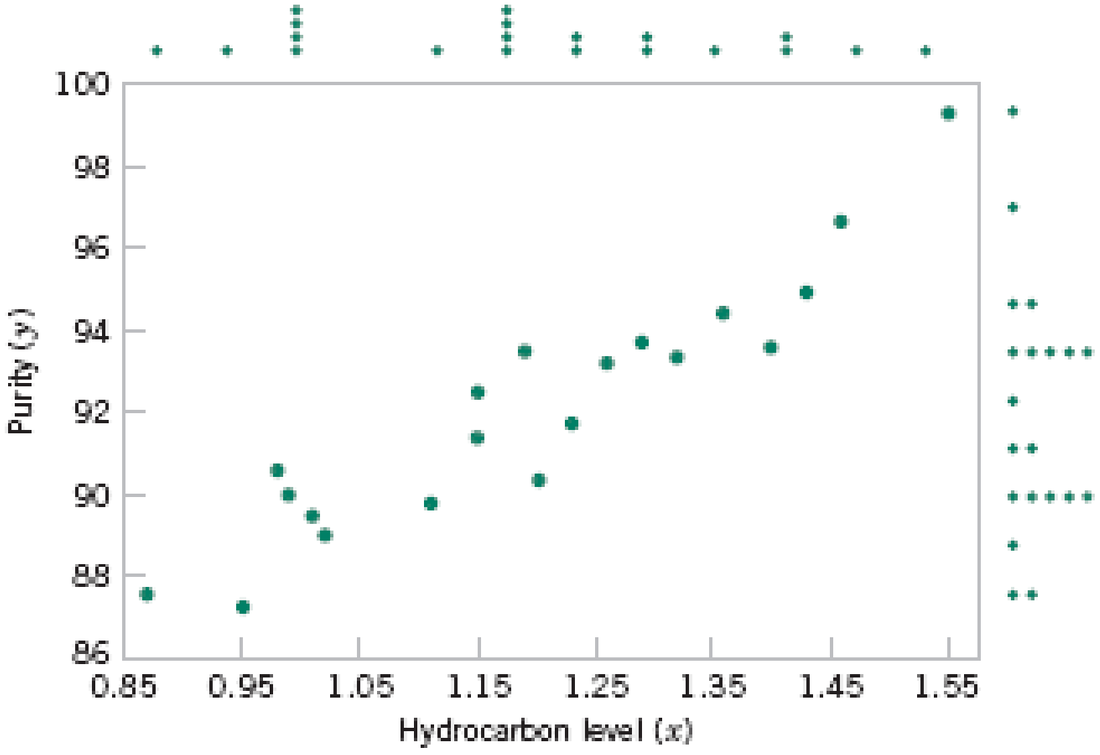
- Apply the `lm()` function.
- Fit a simple linear regression:

```
eruption.lm <- lm(eruptions~waiting, data=faithful)
summary(eruption.lm)
plot(eruptions~waiting, data=faithful,main= "Plot of Y vs X")
abline(eruption.lm, col="red")
```

Linear Regression in R

```
Call:
lm(formula = eruptions ~ waiting, data = faithful)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.29917 -0.37689  0.03508  0.34909  1.19329 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.874016   0.160143  -11.70   <2e-16 ***
waiting      0.075628   0.002219   34.09   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4965 on 270 degrees of freedom
Multiple R-squared:  0.8115,	Adjusted R-squared:  0.8108 
F-statistic:  1162 on 1 and 270 DF,  p-value: < 2.2e-16
```

Example: Oxygen Purity and Hydrocarbon Level



Scatter Diagram of Oxygen Purity Versus Hydrocarbon Levels

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

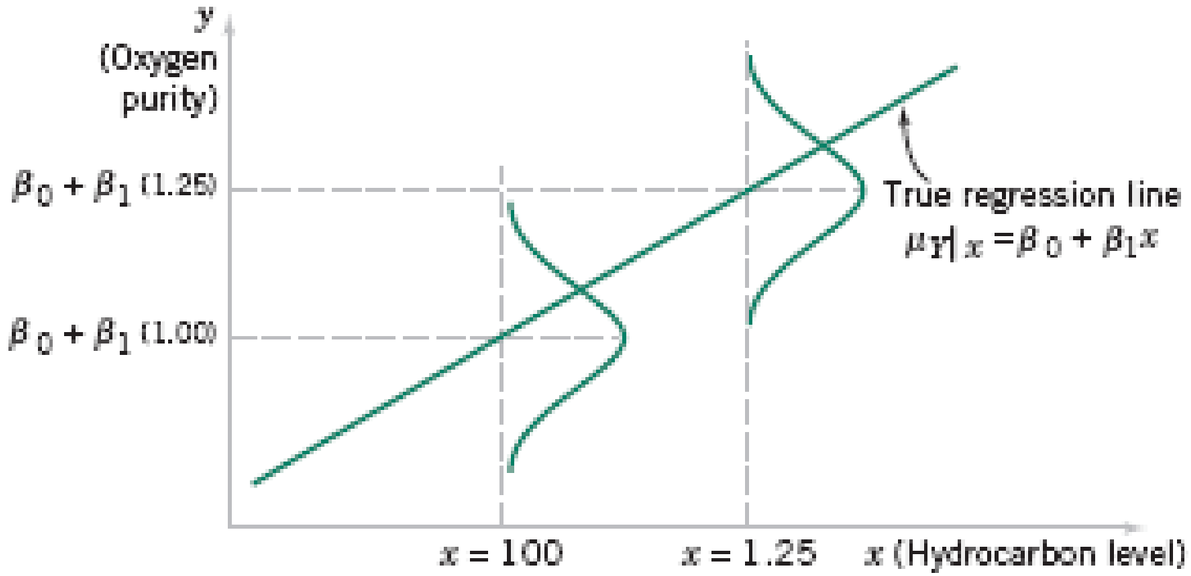
Example: Oxygen Purity and Hydrocarbon Level

```
H20<-c(0.99,1.02,1.15,…,0.95)
O2<-c(90.01,89.05,91.43,…,87.33)
ex<-data.frame(H20,O2)
names(ex)<-c("Hydrogen_Level","Oxygen_Purity")
ex.lm<-lm(Oxygen_Purity~Hydrogen_Level,data=ex)
summary(ex.lm)
plot(Oxygen_Purity~Hydrogen_Level,data=ex,pch=19)
abline(ex.lm)
```

Example: Oxygen Purity and Hydrocarbon Level

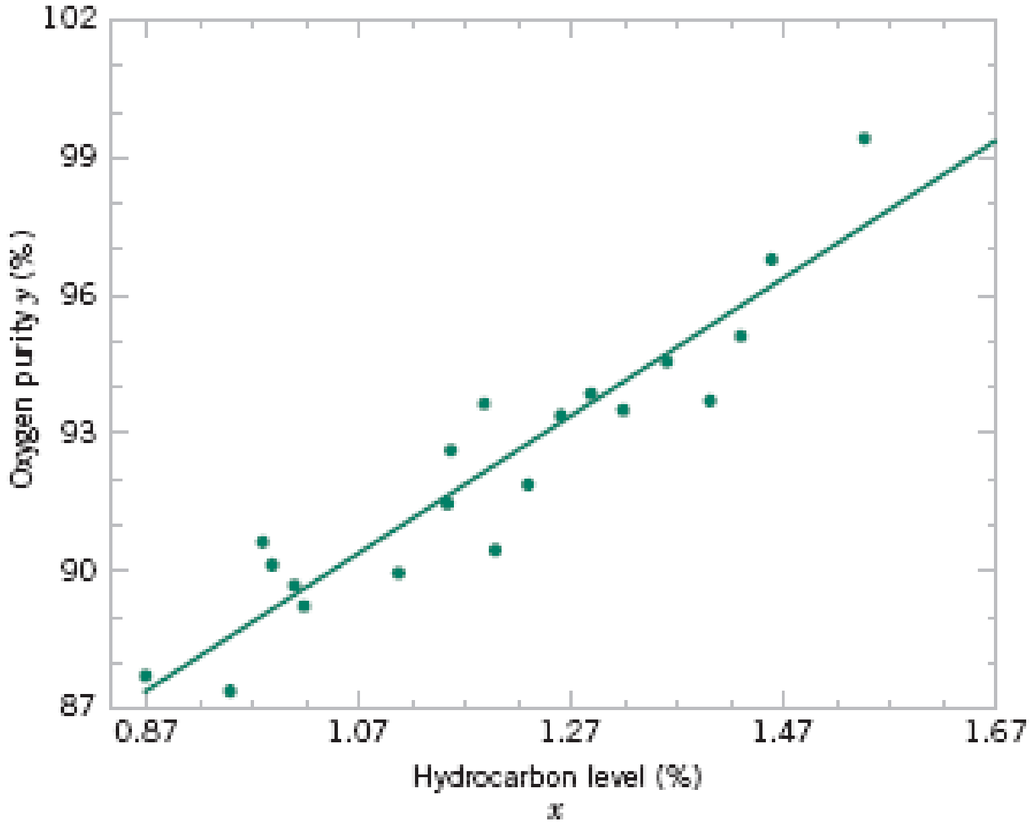
```
Call:
lm(formula = Oxygen_Purity ~ Hydrogen_Level, data = ex)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.83029 -0.73334  0.04497  0.69969  1.96809 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)      74.283      1.593   46.62  < 2e-16 ***
Hydrogen_Level   14.947      1.317   11.35 1.23e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.087 on 18 degrees of freedom
Multiple R-squared:  0.8774,	Adjusted R-squared:  0.8706 
F-statistic: 128.9 on 1 and 18 DF,  p-value: 1.227e-09
```

Example: Oxygen Purity and Hydrocarbon Level



The distribution of Y for a given value of x for the oxygen purity-hydrocarbon data.

Scatter plot of oxygen purity y versus hydrocarbon level x and regression model



Multiple Linear Regression

- Multiple linear regression contains more than one regressor, predictor or independent variable X and a dependent or response variable Y.
- The model for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, k$ where n is the number of observations and k is the number of regressors (independent variable):

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i$$

- The parameter β_j represents the expected change in response y per unit change in x_j when all remaining regressor variables are held constant.

Coefficient of Determination (R^2)

- Coefficient of determination (R^2) measures the proportionate reduction in the total variation of Y associated with the introduction of $X_1, X_2, \dots, X_k$. Hence, the proportion of total variation in Y which have been explained by $X_1, X_2, \dots, X_k$.
- However, R^2 is not suitable to be used to measure the quality of the fit for a multiple linear regression model since it will always increase each time a regressor is added to the model. The increment does not necessarily mean that the model is good.
- The usage of adjusted R^2 (R_A^2) is preferable since its value will only increase if the inclusion of the new variable is able to reduce the Mean Squared Error (MSE).

Linear Regression in R

- **Plot a scatterplot matrix using** `pairs()` **function:**

```
pairs(~stack.loss + Air.Flow + Water.Temp + Acid.Conc.,
data=stackloss)
```

- **Fit a multiple linear regression:**

```
stackloss.lm <- lm(stack.loss~ Air.Flow + Water.Temp +
Acid.Conc., data=stackloss)
summary(stackloss.lm)
```

Linear Regression in R

```
Call:
lm(formula = stack.loss ~ Air.Flow + Water.Temp + Acid.Conc.,
    data = stackloss)

Residuals:
    Min      1Q  Median      3Q     Max
-7.2377 -1.7117 -0.4551  2.3614  5.6978

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -39.9197    11.8960  -3.356  0.00375 **
Air.Flow      0.7156     0.1349   5.307  5.8e-05 ***
Water.Temp    1.2953     0.3680   3.520  0.00263 **
Acid.Conc.   -0.1521     0.1563  -0.973  0.34405
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.243 on 17 degrees of freedom
Multiple R-squared:  0.9136,    Adjusted R-squared:  0.8983
F-statistic:  59.9 on 3 and 17 DF,  p-value: 3.016e-09
```

Dummy-Variable Model

- Involves categorical independent X variables. The qualitative variable X is entered into a regression model through dummy or indicator variables.
- Number of dummy variables is 1 less than the number of levels/categories in the variable.
- It may be combined with quantitative independent X variables.
- The response/dependent variable Y must be quantitative.

Dummy-Variable Model

- If the model only have one quantitative independent variable X_1 and one qualitative variable X_2 with two categories, A and B:

$$x_2 = \begin{cases} 0, & \text{if observation from category A} \\ 1, & \text{if observation from category B} \end{cases}$$

- The model: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$

  For category A: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2(0) + \varepsilon_i = y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$

  For category B: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2(1) + \varepsilon_i = \beta_0 + \beta_1 x_{i1} + \beta_2 + \varepsilon_i$

  β_2 is the average difference in Y between the two categories, when other variables is the same.

Linear Regression in R

- **Fit a dummy-variable model:**

```
class(mtcars$cyl)
mtcars$CYL<-as.factor(mtcars$cyl)
mt.lm<-lm(mpg~CYL,data=mtcars)
summary(mt.lm)
```

Linear Regression in R

```
Call:
lm(formula = mpg ~ CYL, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.2636 -1.8357  0.0286  1.3893  7.2364 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  26.6636     0.9718  27.437  < 2e-16 ***
CYL6         -6.9208     1.5583  -4.441 0.000119 ***
CYL8        -11.5636     1.2986  -8.905 8.57e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.223 on 29 degrees of freedom
Multiple R-squared:  0.7325,	Adjusted R-squared:  0.714 
F-statistic:  39.7 on 2 and 29 DF,  p-value: 4.979e-09
```

Linear Regression in R

Syntax	Model
`Y~A`	$Y = \beta_0 + \beta_1 A$
`Y~-1+A`	$Y = \beta_1 A$
`Y~A+I(A^2)`	$Y = \beta_0 + \beta_1 A + \beta_2 A^2$
`Y~A+B`	$Y = \beta_0 + \beta_1 A + \beta_2 B$
`Y~A:B`	$Y = \beta_0 + \beta_1 AB$
`Y~A*B`	$Y = \beta_0 + \beta_1 A + \beta_2 B + \beta_3 AB$
`Y~(A+B+C)^2`	$Y = \beta_0 + \beta_1 A + \beta_2 B + \beta_3 C + \beta_4 AB + \beta_5 AC + \beta_6 BC$

Residual Analysis

- Residual = Observed value – Fitted (predicted) value
- Measures the error made by the regression model
- Used to check model assumptions and identify problems:
  - Check linearity
  - Check constant variance (homoscedasticity)
  - Check normality of errors
  - Identify outliers and influential observations
- A good model should show random residual patterns

```
res <- residuals(ex.lm)
fitted <- fitted(ex.lm)
```

Residual Analysis: Residuals vs Fitted Values Plot

- Purpose: Detect non-linearity and detect heteroscedasticity (non-constant variance)
- Interpretation:
  - Random scatter around 0 → model is appropriate
  - Curve pattern → non-linear relationship
  - Funnel shape → unequal variance

```
plot(fitted(ex.lm), residuals(ex.lm),
     xlab = "Fitted values", ylab = "Residuals")
abline(h = 0, col = "red")
```

Residual Analysis: Normality of Residuals

- Required for valid hypothesis tests and confidence intervals
- Methods:
  - Histogram of residuals

    ```
    hist(residuals(ex.lm))
    ```

  - Normal Q-Q plot
    - Interpretation: Points close to the line → normality assumption reasonable

    ```
    qqnorm(residuals(ex.lm))
    qqline(residuals(ex.lm))
    ```

Residual Analysis: Standardized vs Studentized Residuals

- Standardized residuals: Residuals scaled by estimated standard deviation

  ```
  rstandard(ex.lm)
  ```

- Studentized residuals: Each residual is scaled using the estimated standard deviation obtained after removing that observation

  ```
  rstudent(ex.lm)
  ```

- |residual| > 2 → possible outlier
- |residual| > 3 → strong outlier

Residual Analysis: Detecting High Leverage Points

- Leverage: Measures how far an observation's predictors (x-values) are from the mean

```
hatvalues(ex.lm)
```

- A value greater than $2(\frac{p}{n})$, where p is the number of predictors (including constant)and n is the number of observations, is large enough to suggest you should examine the corresponding observation.

Residual Analysis: Detecting Influential Observations (DFFITS)

- DFFITS: Measures influence of each observation on fitted values

```
dffits(ex.lm)
```

- A absolute value greater than $2\sqrt{p/n}$, where p is the number of predictors (including constant)and n is the number of observations, is considered large and should be examined.

Residual Analysis: Detecting Influential Observations (Cook's Distance)

- Cook's Distance: Measures the overall influence of an observation on the fitted regression model

```
cooks.distance(ex.lm)
```

- $D_i > 1$ -> potentially influential
- $D_i > 4/n$ -> worth investigating

THANK YOU